Phase Manifold Model: First Principles

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1. Purpose and the FTU Family

PMM proposes an underlying event architecture connecting quantum mechanics (QM), general relativity (GR), and information theory (IT). Its primary goal is that connecting architecture. Its secondary goal is to investigate force manipulation through depth; this is a research objective. Resolving known physics problems is a further possible outcome.

The FTU family gives different readouts of one complete event. Its members describe the event's grades, spatial content, time, mass, energy, action, history and downstream gravitational response. They are not separate events. The central ideas run through the family: **time is angular; space is emergent; depth is different; light is stillness; and M_{op} connects realized relation with motion.**

2. F_{tu} One Complete Event and Contact

$$\mathrm{F}_{tu} \;=\; B \circ D \circ A$$

$$B \;\rightarrow\; @ \;\rightarrow\; D \;\rightarrow\; -@ \;\rightarrow\; A$$

B = Before = Potential → Quantum Mechanics → Wave → Uncertainty → Pre-spatial

D = During = Realization → General Relativity → Particle → Partial Certainty → Spatial xyz

A = After = Transcription → Information Theory → Cube → Full Certainty → Post-spatial

@ = Wave Function Collapse / Contact. It actualizes an admissible pair interaction and opens D. **$-$@ = Anti Contact** marks terminal D closure. C_1 governs entry, C_2 conditions realization, and C_3 governs what persists in A. The QM/GR/IT labels express PMM correspondences; the certainty progression is an event-state heuristic, not complete microscopic knowledge.

Each elementary pair interaction is one shared FTU. Its completion produces two separate endpoint Afters associated with the participant lineages, sharing one event identity and one event-level source energy.

3. The FTU Family at a Glance

Family member	Readout	Role
F_{tu}	B ∘ D ∘ A	Complete event
F^{c}_{tu}	$((c^0)_{mp}, (c^1, c^2)c, (c^3)_{k_B})$	c-order architecture
F^{xy}_{tu}	$(\psi(x, y, z), \mathrm{xyz}, xy)$	Potential, realized and retained spatial content
F^{T}_{tu}	(T^B, T^D, T^A)	Phase-specific temporal readouts
F^{M}_{tu}	$(m_p, \frac{E^D}{c^2}, \frac{\hbar\omega^A}{c^2})$	Mass readouts
F^{E}_{tu}	$(m_p c^2, E^D, \hbar\omega^A)$	Energy readouts
F^{sv}_{tu}	(S^B, S^D, S^A)	Phase-Action Vector
$\mathrm{F}^{\mathrm{led}}_{tu}$	$S^B + S^D + S^A$	Additive Action Ledger
F^{s}_{tu}	$S^B \times S^D \times S^A$	Action Cube
F^{H}_{tu}	(B,(B,D),(B,D,A))	Accumulated phase-history readout
$\mathrm{F}^{\mathrm{Hxy}}_{tu}$	History paired with spatial content	Combined history/spatial readout
F^{G}_{tu}	$\nabla\varphi$	Relational clock-gradient gravity readout

These entries have different units and roles. Definitions organize the architecture; candidate span assignments and the candidate Action Cube retain their stated status. The family is not a single list of independently derived physical laws.

4. F^{c}_{tu} The Ladder of Light Creases and Stillness

The Ladder of Light is PMM's c-order map of one complete event: c^0 represents Before, c^1 and c^2 represent During, and c^3 represents After. The constants m_p, c, and k_B mark their phase creases.

$$c^0 \;\rightarrow\; c^1 \;\rightarrow\; c^2 \;\rightarrow\; c^3$$

$$\mathrm{F}^{c}_{tu} \;=\; ((c^0)_{mp}, (c^1, c^2)c, (c^3)_{k_B})$$

m_p anchors entry, c scales realization, and k_B converts the information readout into entropy units. These constants are phase creases in PMM; their accepted physical values are inputs. Both c^1 and c^2 are D grades, associated with two internal x-y vantages. The native event ladder ends at c^3.

Light as stillness: c is D's invariant still reference. Null propagation has zero proper time without supplying a photon rest frame. This does not remove angular D progression. The scale/action-time identity $\mathrm{L} = cT^{D}_{\mathrm{action}}$ applies on its identified branch; it does not assign every interaction duration or participant separation.

5. F_{tu}^{xy} Emergent Space and Distinct Depth

$$\mathrm{F}_{tu}^{xy} \;=\; (\psi(x,y,z),\ \mathrm{xyz},\ xy)$$

Before B	During D	After A
Potential configuration $\psi(x,y,z)$	Active realized xyz geometry	Retained completed xy readout

B's coordinates describe potential content, not active B-space. Active depth ends at transcription; the retained A readout is information rather than continuing active spatial geometry. A's cube representation is not active three-dimensional A-space.

M_{op} — **Motion Operations Principle:** Space is realized relation, and motion is the evolution of that relation. B carries potential relation and motion; D realizes them; A retains the completed outcome. Its kinematic summary is:

$$M_{op}(\delta_{ij}) \;\equiv\; (\delta_{ij},\ d\delta_{ij}/dt,\ d^2\delta_{ij}/dt^2)$$

Here δ is inter-participant separation and t comparison time. The tuple does not determine the dynamics. An FTU has no isolated physical realization; geometry is relational across events and lineages.

Internal depth z^D arises from the common scale of the two D vantages:

$$\rho_1 \;=\; \sqrt{x_1^D y_1^D},\ \rho_2 \;=\; \sqrt{x_2^D y_2^D}$$

$$z^D \;=\; L \;=\; \sqrt{\rho_1 \rho_2}$$

Depth is distinct from ordinary height and width and from δ. This geometry motivates the depth-and-force objective without establishing a force-control law.

6. F_{tu}^{T} Time Is Angular

$$\mathrm{F}_{tu}^{T} \;=\; (T^B, T^D, T^A)$$

$$T^B \;=\; [\frac{\hbar}{m_p c^2}]\Delta\theta^B$$

$$T^D \;=\; \frac{d}{c}$$

$$T^A \;=\; \frac{\Delta\theta^A}{\omega^A}$$

The current candidate span assignments are $\Delta\theta^B = \arccos\sqrt{P_e@}$ and $\Delta\theta^A = -\ln P_e@$ for $0 < P_e@ \leq 1$. They refer to the same selected contact's original probability and are therefore coupled. The phase-action interpretation of these assignments remains candidate.

Time is angular in PMM, with D progression θ^D running from entry to adopted closure π. Action supplies the common ruler. If $T^D = \mathrm{d/c}$ is identified with $\frac{S^D}{E^D}$, the existing scale relation gives d = L conditionally. Proper time, action time and comparison time retain their separate definitions; the temporal metric is open.

7. F^{M}_{tu} and F^{E}_{tu} Mass and Energy

$$\mathrm{F}^{M}_{tu} \;=\; (m_p,\; \frac{E^D}{c^2},\; \frac{\hbar\omega^A}{c^2})$$

$$\mathrm{F}^{E}_{tu} \;=\; (m_p c^2,\; E^D,\; \hbar\omega^A)$$

The entries have mass and energy units respectively. The adopted D-to-A frequency transcription, together with E $= \hbar\omega$, gives $E^A = E^D$ within one event. Two endpoint records do not create two independent source-energy copies. Neither tuple alone determines the energy selected by an interaction.

8. F^{sv}_{tu} Action as the Common Ruler

$$\mathrm{F}^{sv}_{tu} \;=\; (S^B, S^D, S^A)$$

$$S^B \;=\; \hbar\Delta\theta^B,\; S^D \;=\; \hbar\theta^D,\; S^A \;=\; \hbar\Delta\theta^A$$

Each coordinate has units $\mathrm{J}\cdot\mathrm{s}$. The vector lets PMM describe all three phases with one measure while retaining their different roles. During D:

$$dS^D \;=\; \hbar d\theta^D,\; \theta^D_{\mathrm{cl}} \;=\; \pi,\; S^D_{\mathrm{cl}} \;=\; \pi\hbar$$

c^1 and c^2 carry two internal D vantages, not two independent D action contributions. The same coordinates support two distinct operations: the Ledger and the Cube.

9. $\mathrm{F}^{\mathrm{led}}_{tu}$ The Additive Action Ledger

$$\mathrm{F}^{\mathrm{led}}_{tu} \;=\; S^B \;+\; S^D \;+\; S^A$$

$$\mathrm{F}^{\mathrm{led}}_{tu} \;=\; \hbar(\Delta\theta^B \;+\; \theta^D \;+\; \Delta\theta^A)$$

The Ledger sums phase-action contributions and has units $\mathrm{J}\cdot\mathrm{s}$. At completion:

$$\mathrm{F}^{\mathrm{led}}_{tu,\mathrm{cl}} \;=\; \hbar(\Delta\theta^B \;+\; \pi \;+\; \Delta\theta^A)$$

Version 3.7.5 renames the former additive F^{s}_{tu} to $\mathrm{F}^{\mathrm{led}}_{tu}$. Its original sum-based mathematics is preserved. F^{L}_{tu} is reserved for length and is not the Ledger symbol.

10. F^{s}_{tu} The Candidate Action Cube

$$\mathrm{F}^{s}_{tu} \;=\; S^B \;\times\; S^D \;\times\; S^A$$

$$\mathrm{F}^{s}_{tu} \;=\; \hbar^3\Delta\theta^B\theta^D\Delta\theta^A$$

The Cube is a candidate action-coordinate box product with units $(\mathrm{J}\cdot\mathrm{s})^3$. It is distinct from additive action, the A record cube and ordinary spatial volume. Its three perspectives describe one event:

Perspective	Candidate readout
Before	Prospective completed products for admissible contacts e, using $S_e^D, \mathrm{cl} = \pi\hbar$
During	Selected-event growth with θ^D, holding B and prospective A spans fixed
After	Completed reconstruction at $\theta^D = \pi$

Prospective A spans are not records already transcribed in B or D. For fixed B/A spans:

$$d\mathrm{F}^s_{tu}/d\theta^D \;=\; \hbar^3 \Delta\theta^B \Delta\theta^A$$

At closure:

$$Q(P) \;=\; -\arccos(\sqrt{P})\ \ln\ P$$

$$\mathrm{F}^s_{tu,\mathrm{cl}} \;=\; \pi\hbar^3 Q(P_e@)$$

Q(P) is candidate information-weighted projective contact separation. Both factors depend on the same probability, so the completed product is not a new independent parameter. At P = 1 the Cube vanishes while a contact may complete and the Ledger equals $\pi\hbar$. As $\mathrm{P} \to 0^+$ the candidate product diverges; no divergent observable is asserted. The metric, invariant-volume interpretation and native measurable consequences remain open. Closure count is separate.

11. F^H_{tu} and $\mathrm{F}^{\mathrm{Hxy}}_{tu}$ History and Retained Spatial Content

$$\mathrm{F}^H_{tu} \;=\; (B, (B, D), (B, D, A))$$

This is the current Core's cumulative phase-history readout. It summarizes which phases have occurred; it is not by itself a complete memory encoding or a law describing every transfer of content.

$$\mathrm{F}^{\mathrm{Hxy}}_{tu} \;=\; (B, \psi(x, y, z)) \;\to\; ((B, D), \mathrm{xyz}) \;\to\; ((B, D, A), xy)$$

The combined readout pairs that history with potential, active and retained spatial content. It explains why a completed record can refer to a past realization without keeping its active depth present.

Each participant lineage receives its own endpoint After for event α. The minimal numerical record is $R^A_\alpha = \{\omega^A, a_{\mathrm{cl}}, I^A\}$, with $I^A = -\ln\ P_e@$. Its frequency and terminal planar shape support reconstruction of completed x-y scale and shape. They do not alone establish arbitrary wavefunction, phase or image-sequence reconstruction. Endpoint records accumulate in the respective histories Φ.

12. F^G_{tu} Recursion and Downstream Gravity

The Angular Temporal Recursion Principle, introduced in 3.7.4, postulates:

$$A\alpha(c^3) \;\to\; \Phi \;\to\; B\beta(c^0),\ \text{with } \alpha \text{ causally prior to } \beta$$

Completed history contributes to later possibilities. Coherent contributions can interfere when relative phase is preserved. The native history-to-B preparation map and phase-preservation mechanism remain open; video playback supplies a conditional record-driven example with present apparatus and energy.

The gravity readout belongs to the multi-FTU sector:

$$\kappa \;=\; (\frac{d\theta^D}{dt})/(\frac{d\theta^D}{dt})_0,\ \varphi \;=\; -\ln\kappa$$

$$\mathrm{F}^G_{tu} \;=\; \nabla\varphi,\ g \;=\; c^2\mathrm{F}^G_{tu}$$

κ compares D angular progression rates using the same comparison time. φ is a dimensionless clock potential; its spatial gradient has units m^{-1}, and multiplication by c^2 gives acceleration. Accumulated records condition later realization through this relational clock-response architecture. Source extraction and interaction dynamics retain their stated open status.

13. Proposed Extensions and Reading Guide

Paired temporal/spatial $\mathrm{F}_{tu}^{T,xy}$, spatial-entry F_{tu}^{sr}, distance F_{tu}^{d}, length F_{tu}^{L}, information F_{tu}^{I} and force F_{tu}^{f} have been proposed in separate development requests. They are not adopted here as additional canonical family definitions. The 3.7.5 Core reserves the length symbol and explicitly adopts the Cube/Ledger notation distinction; other extensions require their own integration and precise definitions.

The family supplies a connected account of potential, realization and record. Recovery comparisons and prediction tests remain separate from definitions and adopted principles. No force-control capability or new empirical grade follows from organizing these readouts.

Technical source: PMM 3.7.5 Core v71.35.

Companions: Recovery v1.20 and Predictions v1.25.